

CIRO

Features & Capabilities

Complete feature matrix — all 10 feature areas, every module, every screen

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CIRO (Crisis Intelligence & Response Orchestrator) is a production-ready emergency response platform that connects citizens, field responders, and command centers through a single AI-verified, human-approved pipeline. This document provides a comprehensive breakdown of all implemented features.

Website	https://ciroquick.netlify.app
API Backend	https://bano-qabil-alibabacloud-hackthon-ciro-app-production.up.railway.app
Repository	https://github.com/saqib54/Bano-Qabil-AlibabaCloud-Hackthon-CIRO-APP

This document is part of the CIRO official documentation suite prepared for the Alibaba Cloud x Bano Qabil Hackathon. All content is based on the implemented system.

ToC

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01

Feature Overview at a Glance

Ten feature areas covering the complete emergency response lifecycle

CIRO delivers 10 major feature areas, 34 screens across 3 portals, 82 REST API endpoints, and a deterministic 10-agent AI pipeline that processes every emergency report end-to-end in under 5 seconds. Every feature listed below is live and accessible on the deployed platform.

#	Feature Area	Core Modules	Screens	Status
F-01	Authentication & Identity	Email/Password, Google OAuth, Email OTP, JWT sessions	3	Live
F-02	Emergency Reporting	10 categories, GPS, live camera, voice-to-report, My Reports	2	Live
F-03	AI Auto-Triage Pipeline	10-agent chain: Sentinel, SpamGuard, GeoScout, SatelliteScout, Deep, Corroborator, Verdict, Geo	5	Live
F-04	Command Center (Admin)	AI Decision Center, dispatch, operations map, analytics, broadcasts, live shelters, weather, user mgmt.	11	Live
F-05	Responder Portal (Staff)	Dashboard, assignments, field ops, history, notifications	5	Live
F-06	Citizen Experience	Dashboard, report form, safety map, safe places, alerts, weather, AI assistant, profile	8	Live
F-07	Realtime Engine	WebSocket hub: alert.new, incident.updated, dispatch.assigned, notification.new	1	Live
F-08	Localization & Accessibility	Urdu / English / Roman Urdu, dark mode, system-aware theme	1	Live
F-09	Personalization	Avatars, language preference sync, theme sync, terms consent	1	Live
F-10	Mobile Applications	Android APK (Capacitor 8), iOS project, native Google login, camera, GPS/mic permissions	2	Live

02

Authentication & Identity (F-01)

Three auth modes with JWT rotation, RBAC, and full audit logging

CIRO implements a multi-mode authentication system supporting password-based login, Google OAuth via Firebase, and passwordless email OTP. All sessions are managed with short-lived JWT access tokens (15 minutes) and rotating 7-day refresh tokens. Refresh token reuse triggers automatic family revocation as a security measure.

Method	Flow	Security Controls
Email + Password	bcrypt-hashed password stored at rest; login bcrypt+passwords, Firebase JWT pair	HTTPS, rate limiting on login endpoint
Google Sign-In (Web)	Firebase Auth popup generates an ID token; user authenticates via Google, x509 cert, email, P.SID, etc	Secure cookies, HTTPS, OAuth2
Google Sign-In (APK)	Native account picker plugin (cordova-plugin-googleplus) authenticates via Google, x509 cert, email, P.SID, etc	Secure cookies, HTTPS, OAuth2
Email OTP	6-digit code generated, bcrypt-hashed and stored in Redis; 60-second TTL; 80-second cooldown	Rate limiting, Redis, HTTPS
JWT Sessions	Access token valid 15 minutes; POST /auth/refresh with rotating refresh token (7-day lifetime)	Refresh token rotation, Redis, HTTPS

Role Model: PUBLIC (citizen) | STAFF (responder) | ADMIN (command center dispatcher).

INFO RBAC middleware is applied to every single API route. Refresh token reuse detection and a per-auth-event audit log are active in production.

03

Emergency Reporting (F-02)

60-second multi-modal incident submission with GPS, camera, and voice

The emergency reporting flow is designed to be completed in under 60 seconds. Citizens can report by typing, speaking (Urdu/Roman Urdu/English), or capturing a photo. Every report receives a globally unique incident number and triggers the 10-agent AI triage pipeline automatically.

Capability	Implementation Detail
10 Emergency Categories	Fire, Road Accident, Medical Emergency, Flood, Crime, Gas Leak, Building Collapse, Power Outage, etc.
Incident Numbering	Format: INC-YYYYMMDD-#### (e.g. INC-20260901-0042) — globally unique, traceable throughout lifecycle
14-State Lifecycle	REPORTED -> TRIAGING -> VERIFIED NEEDS_REVIEW REJECTED; VERIFIED -> DISPATCHED RESOLVED
GPS Location	Browser Geolocation API — one-tap capture; manual address fallback with named location storage
Live Camera Capture	getUserMedia() photo stream; capture triggers multipart upload; stored as incident_media records
Voice-to-Report	Web Speech API in ur-PK locale; live interim transcription displayed; AI autofill extracts category, location
My Reports Timeline	Citizens see full status history, current assignment details, ETA, and dispatcher notes; real-time updates
Instant Safety Reply	Authority-approved safety instructions served immediately on submission (never free-form AI output)

04

AI Auto-Triage Pipeline (F-03)

Deterministic 10-agent chain — explainable, auditable, < 5 seconds end-to-end

Every emergency report triggers a deterministic 10-agent pipeline that runs in parallel/sequential stages and completes in under 5 seconds. The pipeline produces a confidence-scored verdict, an impact-zone polygon, a suggested dispatch team, an ETA estimate, and a 2-3 line plain-language copilot summary for the dispatcher. Each agent records its findings, duration, and status in the incident_verification table, creating a full per-report audit trail.

#	Agent	Responsibility	Key Output
1	Sentinel	Ingests the raw report; normalises fields; classifies category and category; outputs a preliminary severity score	category, category, severity_score
2	SpamGuard	Heuristic scoring: short description +30, no phones +10, no excess, spam, <150% actions, title == description	spam_score
3	GeoScout	Validates coordinates within Pakistan bounding box; performs reverse geocoding; coordinate validation	location, name, reverse_geocode, coordinate_validation
4	SatelliteScout	Deterministic coordinate-hash simulation of FIRMSatellite signals; satellite signal status, confidence, and landslide risk	satellite_signals, confidence, landslide_risk
5	DedupGuard	Queries same category within approx 1 km radius duplicate flagging; duplicate flagging	SUSPECTED_DUPLICATE
6	Corroborator	Counts independent nearby reports for the same category within the same time window; confidence boost	nearby_reports, confidence_boost
7	VerdictEngine	Combines all agent signals into a composite verdict; verdict: AUTO_VERIFIED, NEED_REVIEW, SUSPECTED	verdict
8	GeoImpact	Computes impact-zone geometry: sector (FIRE/GAS/Earthquake), impact radius, corridor (ACSD/HR/Police)	impact_zone, corridor
9	SmartDispatch	Matches category to TEAM_MATRIX (e.g., ACCIDENT_Triage, Ambulance, Traffic Police); estimates ETA = distance/speed	dispatch_team, eta
10	Copilot	Generates a 2-3 line plain-language summary for the dispatcher; highlights key facts: type, location, severity	summary

NOTE Human Approval Gate: NO team is ever dispatched by AI alone. A dispatcher must click 'Approve & Dispatch' — every decision is audit-logged with actor ID, timestamp, and before/after state.

Auto-actions (safety-gated): Only when verdict = AUTO_VERIFIED AND severity >= HIGH AND spam_score < 40 will the system automatically issue a public alert and pre-route to the department. All other cases require dispatcher decision.

05

Command Center — Admin Portal (F-04)

11 screens, full operational control, AI-assisted decision making

The Admin portal is the operational nerve centre of CIRO. Dispatchers see confidence-scored AI verdicts, approve dispatch, manage resources, issue emergency broadcasts, and monitor analytics — all in real time.

Screen	Purpose & Key Features
Admin Dashboard	Live KPI cards: active incidents, average response time, department load, resolved today. Quick-action
AI Decision Center	Three queues: NEEDS_REVIEW (SpamGuard flagged), LOW_CONFIDENCE (VerdictEngine below th
Smart Dispatch	Team assignment with SmartDispatch recommendations. Shows available staff by department and ET
Operations Map	Live severity-pulsed incident pins, impact-zone polygons (sector/corridor/circle/private), forecast hotsp
Analytics Dashboard	Trend line charts, category breakdown pie, region heatmap, response-time distributions, auto-verified
Emergency Broadcasts	Compose area-targeted alerts with severity level and target audience. Fan-out: DB notification records
Shelters & Safe Places	Full CRUD registry: name, type, address, coordinates, capacity, contact, active toggle. Citizens see th
Weather Center	Point-Forecast API integration: current conditions, 15-day outlook, hourly data, AQI, advisories. Risk-a
Staff / User / Department Management	Full CRUD for users (role, status, activation). Department management with contact info and active tog
Audit Logs	Tamper-evident trail of every privileged action: dispatch approvals, rejections, user management, settin
Resolution Review	Before/after evidence review workflow for closing incidents. Approve, reject, or reopen resolutions sub

06

Responder Portal — Staff (F-05)

Verified, de-duplicated assignments with full field operation support

Screen	Purpose
Staff Dashboard	Today's assignment count, severity breakdown, duty status toggle, quick navigation
Assignments	List of verified, de-duplicated tasks with location pin, category, severity, people affected, copilot summ
Incident Operations	Accept assignment; update field status (en-route, on-scene); add situation logs with photos; submit res
History	Completed responses per responder with timestamps, outcome, and evidence links
Notifications	Real-time dispatch push notifications and emergency broadcast feed

Seeded departments: Rescue 1122, Fire Brigade, Traffic Police, Police, WAPDA. SmartDispatch routes assignments to the most appropriate department based on emergency category, nearest ON_DUTY staff, and ETA estimate.

07

Citizen Experience (F-06)

8 screens delivering safety, transparency, and real-time feedback

Screen	Purpose & Detail
Public Dashboard	Live alert ticker showing active verified incidents; nearby incident count; quick-action buttons for Report
Report Emergency	The core 60-second flow: 10-category selector, voice button (Urdu/Roman Urdu/English), live camera, and
Safety Map	Leaflet map with severity-pulsed incident pins (color-coded: red=critical, orange=high, yellow=medium, green=low)
Safe Places	Directory of shelters with name, type, address, capacity, contact number, and one-tap directions. Manual
Alerts	Emergency broadcast feed from command center. Real-time WebSocket updates; severity color coding
Weather	Hyperlocal weather forecast with current conditions, 15-day outlook, AQI, hourly breakdown, and safety
AI Safety Assistant	Safety Q&A chatbot with authority-approved response templates. Never free-form AI output. Supports
My Reports	Personal incident timeline: status history, dispatcher notes, ETA, assigned team. Real-time WebSoc
Profile	Avatar upload, full name edit, language selection (Urdu/English/Roman Urdu), theme toggle (light/dark)

KEY POINT Medical Privacy by Design: medical incident impact zones are visible to assigned teams only. Affected population count is stored as 0 for medical incidents. The public map never exposes medical emergency details.

08-10

Cross-Cutting Features

Realtime engine, localization, personalization, mobile apps

Area	Detail
Realtime (F-07)	WebSocket hub using the ws library. Events: alert.new, incident.updated, dispatch.assigned, notification
Localization (F-08)	Full i18n: Urdu, English, Roman Urdu. Runtime language switch without page reload. Language prefer
Dark Mode (F-08)	System-aware dark mode detection + manual toggle. Design tokens: navy #0A1E42, periwinkle #D7E3F
Personalization (F-09)	Avatar upload (multipart, server-side storage), language & theme preferences synced to user record o
Android APK (F-10)	Capacitor 8 build from the same React codebase. Camera, GPS, and microphone permissions configu
iOS Project (F-10)	Capacitor iOS project at client/ios/ with usage-description plist entries for camera, microphone, and loc

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Scale, Capacity & AI Feature Matrix

Metric	Value	Notes
REST API endpoints	82	Across 11 route modules
Total screens	34	Across 3 portals (citizen/responder/admin)
AI pipeline latency	< 5 s (p95)	2-20 ms in offline fallback mode
WebSocket push latency	< 1 s	< 300 ms measured
Extract API latency	< 500 ms	~5 ms offline fallback
Auth modes	3	Password, Google OAuth, Email OTP
Languages	3	Urdu, English, Roman Urdu
Database tables	18+	Including 10 migration files
Deployment targets	3	Web (Netlify), API (Railway), APK + iOS
Emergency categories	10	With tailored AI handling per category
AI agents per report	10	Deterministic, explainable, auditable

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AI Feature Matrix — All AI Capabilities

AI Feature	Behaviour / Implementation
Smart Report Detection	Classifies Fire / Medical / Accident / Crime / Flood and 5 more from text, voice, or photo via Qwen
Auto Location Detection	GPS coordinates + description -> named location using Pakistani cities/landmarks vocabulary;
AI Risk Scoring	Four levels: Critical / High / Medium / Low with confidence percentage; composite from 10 agents
Voice-to-Report	Urdu, English, Roman Urdu -> auto-filled form: category, location, people, priority, suggested text
Duplicate Detection	DedupGuard: same category within ~1 km and 24 h -> merge flag with linked master incident ID
Fake/Spam Detection	SpamGuard heuristics: score >= 40 gates auto-actions; >= 60 forces human review; findings listed
Satellite Signal Fusion	SatelliteScout: FIRMS/Sentinel-1 simulation for fire, flood, landslide; SUPPORTING/INCONCLUSIVE
Corroborating Reports	Corroborator: independent nearby reports RAISE confidence (unlike duplicates which flag suspicious)
Impact Zone Geometry	GeoImpact: sector (fire/gas), corridor (accident/flood), private circle (medical), general circle —
Instant AI Response	Authority-approved safety instructions per category returned immediately on submit (never freeform)
Smart Dispatch	SmartDispatch: TEAM_MATRIX per category; ETA = clamp(4 + distance/800m [+2 if CRITICAL])
Emergency Forecasting	Historical clustering on 2 km grid; risk = min(100, count*14 + severity_weight*6); rendered as map
Command Center Copilot	Per-report 3-line plain-language summary: type, location, confidence, satellite signal, affected, suggested

AI Feature	Behaviour / Implementation
Weather Intelligence	Qwen meteorologist briefing: current, 15-day, hourly, AQI, lifestyle insights, radar coordinates; s